

# Falak Rana

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## TECHNICAL SKILLS

**Programming:** Python, JavaScript, TypeScript, Rust, Java, C#

**Frontend:** React.js, Next.js, Angular, HTML5, CSS3, Tailwind CSS

**Backend:** FastAPI, .NET, Flask, Springboot, Node.js, Express.js, REST APIs, Microservices Architecture

**AI / ML:** Supervised Learning, Deep Learning, PyTorch, TensorFlow, LangChain, Retrieval-Augmented Generation (RAG)

**Databases:** MongoDB, MySQL, PostgreSQL, SQLite, Redis

**DevOps & Tools:** Git, GitHub, Docker, Linux, Postman, AWS (EC2, S3, CloudWatch)

## EXPERIENCE

### Software Development Intern

Jan 2026 – Present

*Infodesk India Private Limited (Onsite)*

Developing scalable microservices and REST APIs using **.NET (C#)** and **Spring Boot (Java)** for enterprise systems

Building responsive UIs with **React**, **Next.js**, **TypeScript** and managing async workflows using **Redux Saga**

Managing cloud infrastructure on **AWS** (EC2, S3, CloudWatch) for automation, logging, and deployment

Designing and optimizing databases using **SQL** and **MongoDB**

Developing high-performance **Rust** CLI tools for legal document analysis

### Machine Learning Intern

Jan 2025 – Apr 2025

*UnifiedMentor (Remote)*

Delivered 6 client-ready ML applications including prediction systems and **AI APIs**.

Built **Flask-based AI services** to expose trained models for real-world use.

Optimized models for better accuracy and faster inference.

## PROJECTS

### CacheFlux

*Python, FastAPI, Redis, PostgreSQL, Docker*

Designed a distributed caching layer using Redis over PostgreSQL, significantly improving API response time under high-traffic workloads.

Implemented cache-aside pattern with TTL and invalidation strategies; containerized using Docker for scalable deployment.

### ChronoNote

*SpringBoot, PostgreSQL, React, Docker*

Built a structured note-taking platform with timeline-based organization, enabling users to capture, search, and manage notes efficiently.

Implemented secure CRUD APIs, fast keyword filtering, and containerized deployment workflow for consistent development and production environments.

### Coursera Main Page Assistant

*Python, RAG, Playwright*

Developed a browser extension leveraging Retrieval-Augmented Generation (RAG) to answer course-specific queries contextually.

Integrated Gemini API for real-time responses, enhancing learning experience with accurate and contextual assistance.

## EDUCATION

### Parul University

*B.Tech in Computer Science Engineering*

2022 – 2026

CGPA: 8.28/10.00

### Green Valley High School

*12th Grade*

2022

91.80%

## CERTIFICATIONS

• **Data Science Bootcamp** (Udemy)

• **Linux Command Line & Shell Scripting** (Cisco)

• **Tableau for Data Visualization** (Udemy)

• **AWS Cloud Practitioner** (AWS Cloud)